

Part C – BIOS vs UEFI Comparison

Feature	BIOS	UEFI
Definition	Older firmware interface used to initialize hardware and start the OS	Modern firmware interface that initializes hardware and starts the OS
Boot Process	Uses traditional BIOS boot process	Uses EFI boot files stored in the EFI System Partition
Maximum Disk Support	Commonly associated with MBR and disks up to about 2 TB	Supports GPT and very large disks
Partition Style	MBR	GPT
Security	Limited built-in boot security	Supports features such as Secure Boot
Speed	Generally slower boot process	Generally faster and more efficient
Advantages	Simple and compatible with older systems	Faster, modern, secure, supports larger disks
Disadvantages	Older technology and more limitations	May have compatibility issues with very old hardware/software
Modern Usage	Mostly legacy systems	Standard on modern computers

UEFI (Unified Extensible Firmware Interface) has largely replaced BIOS (Basic Input/Output System) because it provides features that are more suitable for modern computers. Both BIOS and UEFI have an important role during startup because they initialize the computer's hardware and help start the operating system. However, BIOS is an older technology with limitations that make it less suitable for newer hardware and operating systems.

One major advantage of UEFI is its support for larger storage devices. Traditional BIOS commonly works with the MBR (Master Boot Record) partitioning system, which has limitations such as supporting disks only up to around 2 TB and allowing only a limited number of primary partitions. UEFI commonly uses GPT (GUID Partition Table), which supports much larger storage devices and more partitions. This is important today because computers and servers can use storage drives with capacities greater than 2 TB.

UEFI can also provide a faster and more efficient boot process. Instead of relying on the older BIOS boot method, UEFI can directly work with boot files stored in an EFI System Partition. This allows modern operating systems to start more efficiently and gives manufacturers more flexibility when designing computer systems.

Another important reason UEFI is preferred is security. UEFI supports a feature called Secure Boot, which can help prevent unauthorized or malicious software from loading during the startup process. This provides an additional layer of protection before the operating system fully starts. Traditional BIOS does not provide the same modern Secure Boot capability.

UEFI is also designed to work better with newer hardware and operating systems. BIOS was developed when computers were much simpler, so some of its design limitations are no longer practical for today's systems. UEFI provides a more modern firmware environment that can support newer technologies while still performing the basic task of preparing the computer before the operating system loads.

Despite these advantages, BIOS was very important in the development of personal computers and is still found in older systems. It is simple and compatible with many older devices and operating systems. However, its limitations in storage capacity, security, and modern hardware support have made UEFI a better choice for current computers.

In conclusion, UEFI has largely replaced BIOS because it offers better support for large storage devices, GPT partitioning, improved security through Secure Boot, and a more modern boot environment. Learning the difference between BIOS and UEFI is especially important in system administration because administrators need to understand how computers and servers start before the operating system takes control. UEFI provides the features needed by modern computer systems, which is why it has become the standard firmware technology used today.